Itauma Itauma, PhD, MBA

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Associate Professor specializing in data analytics, artificial intelligence, and applied machine learning.
Extensive experience leading graduate programs, designing industry-aligned curricula, and mentoring doctoral research in analytics and AI. Proven record of integrating tools such as Python, Power BI, and simulation platforms into applied learning environments. Active researcher with publications in AI, analytics, and digital health, contributing to national and international conferences.

Experience

Associate Professor of Data Science and Analytics, 2026 – Present

Harrisburg University of Science & Technology, Harrisburg, PA

- Develop and teach graduate courses in data science, analytics, machine learning, statistical modeling, and applied research.
- Mentor master's and doctoral students in research design, data analysis, scholarly writing, and dissertation development.
- Design industry-aligned curricula and experiential projects using Python, R, SQL, Power BI, and real-world datasets.
- Maintain an active research agenda and collaborate with faculty and students on publications, conference presentations, and applied analytics projects.
- Support program assessment, curriculum improvement, student advising, and departmental service.

Faculty/Division Chair/Assistant Professor of Analytics, 2020 – 2026

Northwood University, Midland, MI

- Led MSBA and DBA programs, overseeing curriculum strategy, faculty coordination, and program assessment
- Designed and delivered graduate and doctoral courses in AI, machine learning, and business analytics
- Chaired doctoral dissertations, guiding research design, statistical modeling, and publication development
- Spearheaded curriculum innovation integrating Python, R, SQL, and Power BI
- Mentored student research leading to peer-reviewed conference publications
- Led cross-functional initiatives aligning analytics education with industry and emerging AI trends

Data Science Consultant, 2025

LearningMate, Remote

- Developed AI-enhanced digital learning and analytics content, applying structured data workflows and content optimization methods to support scalable, high-quality delivery.
- Collaborated with cross-functional teams to design, review, and deliver analytics-focused digital modules aligned with learner and stakeholder needs.

Faculty, Department of Computer Science, 2016 – 2020

Wayne State University, Detroit, MI

• Taught undergraduate and graduate courses in data analytics, programming, and computer science including Python, R, and machine learning.

• Developed instructional strategies and created engaging course content for data analytics and programming courses.

• Mentored students on quantitative research projects, leading to conference publications in data analytics and machine learning.

Faculty / Technical Program Facilitator, Data Analytics/Data Science, 2016 – 2019

Southern New Hampshire University, Remote

• Facilitated and taught courses in data analytics, data science, and computer science for the STEM program, with a focus on data-driven decision-making and analytics tools.

• Led program development and course design, serving as a subject matter expert in data analytics tools such as Python, R, SQL, and Power BI.

• Contributed to the development of competency-based learning models, ensuring courses met industry standards and academic rigor.

Graduate Teaching & Research Assistant, Department of Computer Science, 2013 – 2015

Wayne State University, Detroit, MI

• Conducted research utilizing high-performance computing cluster (HPCC) to solve big data problems, including image analysis using machine learning algorithms.

• Provided instruction and assessment support for computer science courses and laboratory sessions, ensuring academic rigor and student success.

Education

Data Science Fellow, 2025

University of Pennsylvannia Center for Learning Analytics

Certificate in Data Science Methods for Digital Learning Platforms.

Program supported by the Institute of Education Sciences, U.S. Department of Education to the University of Pennsylvania.

Executive Master of Business Administration, 2025

Quantic School of Business and Technology

Specializations: Strategic Leadership, Advanced Statistical Inference

Harvard Business Analytics Program, 2020

Harvard University

Executive Graduate Certificate in Business Analytics

• Relevant coursework: Digital Strategy, Quantitative Analysis, Leadership, Data Science Systems, Data-Driven Marketing

Ph.D. in Instructional Design and Technology, 2019

Keiser University

Dissertation: Pre-College Factors That Predict Intentions of Minority Females to Enroll in College STEM Programs

• Graduated with 4.0 GPA

Master of Science in Computer Science, 2015

Wayne State University

Focus: Big Data Analytics

Thesis: Unsupervised Learning and Image Classification in High-Performance Computing Clusters

Master of Science in Computer Engineering, 2012

Istanbul Technical University

Focus: Human-Robot Interaction

Thesis: Gesture Imitation Learning in Human-Robot Interaction

B.Eng in Electrical Engineering, 2004

University of Ilorin

Area of Concentration: Computer and Control Engineering

Thesis: Microprocessor Driven Traffic Light for a Dual Cross-Junction Road

Skills

Artificial Intelligence, Business Intelligence, Analytics, Data Modeling, Statistical Analysis, Machine Learning, Programming, Predictive Analytics, Python, R, SQL, SAS, Power BI, Data Visualization, Databases, Course Development, Instructional Design, Faculty Development, Critical Thinking, Mentoring, Curriculum Development, Program Leadership

Links

LinkedIn: linkedin.com, GitHub: github.com

Scholarly Publications & Research Contributions

Peer-Reviewed Journal Articles

Itauma, I., & Itauma, O. (2026). Sleep Quality and Daytime Functioning as Predictors of Metabolic Risk: A Structural Equation Modeling Analysis of NHANES Data. International Journal of Digital Health and Telemedicine, *2*(1). https://doi.org/10.51137/wrp.ijdht.782

Srikanth, S., & Itauma, I. (2026). Leveraging Generative AI as a Socio-Technical Mediator in Global Project Communication and Conflict Management. Paper accepted at the 3rd International Business Analytics Conference, Fredonia, NY.

Srikanth, S., & Itauma, I. (2025). Transforming Project Management through Data Analytics and Generative AI. Empirical Economics Letters, 24(September Special Issue 1), 93–109. https://doi.org/10.5281/zenodo.18866470

Itauma O., Itauma I., 2024. AI Scribes: Boosting Physician Efficiency in Clinical Documentation. International Journal on Bioinformatics & Biosciences. 2024 Mar;14(1):09-18. 10.5121/ijbb.2024.14102.

Itauma O., Abara C., Gonullu D., Fee M., Itauma I., Awonuga A., 2020. Contraceptive choices by weight status among women in the United States: An analysis of the 2015-2017 National Survey of Family Growth. Fertility and Sterility, 114(3), pp.e173-e174.

Itauma I.I., Kivrak H., and Kose H., 2012: Gesture imitation using machine learning techniques. 20th Signal Processing and Communications Applications Conference (IEEE SIU'12), pp. 1-4, April 18-20. Mugla, Turkey.

Kose-Bagci H., Yorganci R., Itauma I.I., 2011: Humanoid Robot Assisted Interactive Sign Language Tutoring Game. 2011 IEEE International Conference on Robotics and Biomimetics, pp. 2247-2248, Dec. 7-11. Phuket Island, Thailand.

Conference Proceedings & Presentations

Ravi, M., Yenuga, H., Itauma, I. (2025). Revolutionizing recruitment: Enhanced machine learning models for bias mitigation and efficiency. Proceedings of the 2nd International Business Analytics Conference, Fredonia, NY.

Rawat, P., Sharma, A., Itauma, I. (2025). Retaining value: Segment-based predictive models for customer churn in banking. Proceedings of the 2nd International Business Analytics Conference, Fredonia, NY. [Best Paper Award]

Itauma, I. (2024). Leveraging large language models for predictive analytics and code generation using ChatGPT and Gemini API. Proceedings of the 1st International Business Analytics Conference, Fredonia, NY.

Mutha, R., Itauma, I. (2024). Machine learning approaches for predicting wine quality from chemical properties in Azure Machine Learning Studio. Proceedings of the 1st International Business Analytics Conference, Fredonia, NY.

Nasim, M., Zhu, S., Itauma, I. (2024). An empirical analysis of the functionalities and confidence scoring mechanisms in leading large language models. Proceedings of the 1st International Business Analytics Conference, Fredonia, NY.

Itauma I., 2024. Technology Integration for Diverse Learners: Enhancing Engagement and Accessibility. Regional Center for Educational Planning (RCEP) UNESCO Symposium.

Itauma I., Robertson A., Fuda-Daddio J., Komaroff E., 2024. Factors Predicting Mathematics and Science Motivation among Minority Female High School Students. American Educational Research Association (AERA) 2024 Annual Meeting, Philadelphia, PA.

Omunguye D., Joshua Hopkins J., Guerrero A., Itauma I., 2023. Assessing ML Algorithms for VTE Risk Classification. Poster presented at the AI in Health Conference, Houston, TX

Renz J., Itauma I., Cutt K., 2020. COVID-19 Tracking App: An HBAP Coronal Project. Harvard Business Analytics Program.

Itauma I., 2020. Data Management & Analytics Simplified For The New Normal In The Health System. Hospital Data Management & Analytics Conference.

Itauma I., 2020. Engaging Students With Canvas in Online and Traditional Classrooms. Michigan Canvas Users Conference.

Messner A. and Itauma I., 2019: Exploring Co-Occurring Mental Illness and Substance Use Disorder Using HPCC Systems. Poster presented at the HPCC Systems Engineering Summit - Community Day.

Itauma I., 2018: Conducting Exploratory Data Analysis in Educational Research Using HPCC Systems. HPCC Systems Engineering Summit - Community Day.

Itauma O. and Itauma I., 2018: Cervical Cancer Risk Factors: Exploratory Analysis using HPCC Systems. HPCC Systems Engineering Summit - Community Day.

Walker C. and Itauma I., 2017: Is the secret to longevity eating chocolate? HPCC Systems Engineering Summit - Community Day.

Itauma I. and Chen X., 2016: Unsupervised Learning and Image Classification in High Performance Computing Cluster. HPCC Systems Engineering Summit Community Day (Poster Presentation & Speaker).

Itauma I., Bubb T., 2016: Academic Success Centers: Improving Resource Utilization using Grounded Theory Methodology. The 15th Annual South Florida Education Research Conference, June 4. Miami, Florida, USA.

Itauma I., Aslan M.S, Villanustre F., and Chen X., 2015: Unsupervised Learning and Image Classification in High Performance Computing Cluster. 14th International Conference on Machine Learning and Applications (IEEE ICMLA'15), December 9-11. Miami, Florida, USA.

Itauma I.I., Kose-Bagci H., 2012: Gesture Imitation Learning for Human-Robot Interaction. IK 2012 (Emotion and Aesthetics) conference, March 16-23. Gunne, Lake Mohne, Germany.

Books & Scholarly Contributions

Itauma I., 2024. Machine Learning Using Python. Quarto Publishing.

Itauma I., Aslan M. S., Chen X. W., Villanustre F., 2016: Unsupervised Learning and Image Classification in High Performance Computing Cluster. Big Data Technologies and Applications, pp.387-400, Springer International (Book Chapter).

Academic Service & Leadership

- Chair, AERA Annual Meeting (2026)
  Session: Teacher Leadership, Wellbeing, and Retention
- Discussant, AERA Annual Meeting (2026)
  Session: Preparing Future Educators in the Age of AI
- Moderator, 1st International Business Analytics Conference, Fredonia, NY (2024)
  Track: Analytics in Business Education
- Dissertation Chair, DeVos Graduate School, Northwood University (2023–Present)

Awards & Recognitions

- Learning Analytics in STEM Education Research (LASER) Scholar, funded by the National Science Foundation — 2023
- Certificate of Appreciation, Northwood University — 2023
- Harvard Business Analytics Program Certificate of Excellence — 2020
- HPCC Systems Community Innovator — 2015-2020
- Best Reviewer, CS Student Workshop, Istanbul — 2011
- Bilateral Education Scholarship, Nigerian Govt. — 2007–2012

Teaching Portfolio (Selected Courses & Tools)

Simulation & Optimization

- **DSE 5070 – Introduction to Data Computing and Programming**
  Monte Carlo simulation, stochastic modeling, decision analysis (Python)
- **MGT-676 – Integration & Implementation (MBA)**
  Strategic decision-making using **CAPSIM simulation**, scenario planning, operational optimization

Business Intelligence & Data Visualization

- **DAT-210 – Foundations of Data Analytics**
  Data wrangling, exploratory analysis, business reporting
- **DAT-410 – Decision Support Presentation**
  Executive dashboards using **Power BI and Tableau**
- **HR-632 – HR Analytics & Technology**
  Workforce analytics and predictive modeling using **Power BI**

Design & Development of Analytical Systems

- **MGT-756 – Business Process Analytics**
  Process modeling, Six Sigma, system optimization
- **MTH-752 – Applied Data Analytics**
  End-to-end analytics pipeline development
- **MGT-683 – MSBA Capstone**
  Full lifecycle analytics system design using real-world datasets

Artificial Intelligence for Business

- **MGT-661 – Artificial Intelligence & Business Analytics**
  NLP, deep learning, recommendation systems
- **MGT-665 – Solving Problems with Machine Learning**
  Classification, clustering, regression
- **MGT-663 – Forecasting & Predictive Analytics**
  Time series and predictive modeling

Programming & Analytical Tools Integrated Across Courses

- Python, R, SQL, SAS
- **Power BI (dashboarding & business intelligence)**
- **CAPSIM (strategic business simulation)**
- Azure Machine Learning, GitHub, Jupyter Notebooks

Dissertation Chairing & Supervision

DevOs Graduate School of Management, Northwood University

2023 – Present

- Supervised doctoral candidates from proposal through successful defense.
- Advised on methodology, statistical tools (SAS, R, SPSS), literature reviews, and compliance with academic standards.

Languages

English *(Native)*, Turkish *(Highly proficient)*, German *(Proficient)*